RETAILSPHERE (PTY) LTD

Data Platform Programme

# Data Integration Design

*Step 10 Deliverable — Data Integration Design*

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# 1. Purpose

This document defines the pipeline architecture pattern for ingesting, transforming, and loading data across RetailSphere's Bronze, Silver, and Gold layers, operationalizing the orchestration tool decision made in ADR-005 (Azure Data Factory). It is intended for engineers building the pipelines in Steps 13–16, and satisfies the reconciliation, error handling, data quality, and POPIA propagation requirements defined in FRD-001 (FR-CCR-005 through FR-CCR-008).

# 2. Scope

DID-001 defines the general Bronze/Silver/Gold pipeline pattern applicable across all six business domains. It does not repeat the tool decision already locked in ADR-005, nor does it fully detail domain-specific transformation logic for domains where STM-001 field-level mapping remains deferred to build phase (Inventory, Procurement, CRM/Marketing, Finance) — only Sales domain currently has confirmed mappings. Domain-specific pipeline detail for the remaining five domains will be finalized immediately prior to each domain's build sprint, consistent with STM-001's own deferral pattern.

Three source systems require deviation from the standard pipeline pattern and are addressed explicitly in Section 3: Meta and GA4 (both arrive 24–48 hours late by design), Supplier Contract PDFs (LLM-assisted extraction, not a standard ADF connector), and Legacy SQL Server DW (discovery-and-decommission target, not an ongoing pipeline source).

# 3. Bronze Layer Ingestion Pattern

Bronze is the raw landing zone for all 10 source systems: data lands exactly as received, with no transformation, preserving full reprocessing capability without needing to re-extract from source.

**Load pattern:** Given RetailSphere is a new platform, all sources begin with a full initial load. Ongoing ingestion proceeds incrementally where the source system supports change detection (e.g. CDC, delta timestamps); sources without this capability (see SSI-001 per-system detail) fall back to full re-extraction on each run.

**Late-arriving sources (Meta, GA4):** No special Bronze-layer handling is required. The 24–48 hour confirmation lag is a characteristic of these platforms' own APIs, not a pipeline failure — Bronze lands whatever confirmed data each source returns on its normal schedule. The lag is not a Bronze-layer risk; it is a downstream Silver/Gold-layer risk (FactContactHistory's contact-count accuracy against the POPIA 4-contact monthly cap), already addressed via the optimistic-count-plus-breach-alert approach agreed with Priya's team (SSI-001).

> ***Resolved: Meta and GA4*** *ingestion runs on a dedicated schedule of every 4 hours (6 runs/day), distinct from the platform's standard once-nightly batch used elsewhere. This is required specifically to satisfy FR-CUS-008's same-day POPIA breach alert commitment — a once-nightly schedule cannot reliably catch a confirmation that lands at an arbitrary time within the day. All other sources remain on the standard nightly schedule; this exception applies only to Meta and GA4.*

**Supplier Contract PDFs:** Raw PDF files land in Bronze unchanged, consistent with every other source. The pypdf-extraction-and-LLM-structuring pipeline runs as a **Silver-layer transformation**, not a Bronze-layer step — structuring unstructured text into defined fields (contracted rate, lead time, payment terms) is genuine transformation work, and treating it as such keeps Bronze's "raw, as received, zero transformation" rule consistent across all 10 sources without exception. This also means the extraction can be safely rerun against the Bronze-stored PDFs if the LLM structuring logic changes, without needing to re-source the original documents.

**Legacy SQL Server DW:** Not an ongoing Bronze ingestion source. Per SSI-001's discovery-and-decommission approach, this system is out of scope for DID-001's ongoing pipeline design; its two undocumented reports will be replicated as part of the Legacy DW discovery workstream, independent of the standard Bronze pattern.

## 4. Silver Layer Transformation Pattern

Silver is where raw Bronze data becomes cleaned, conformed, and trustworthy — cleansing, standardization, and the two identity resolution processes (ADR-001 for Product, ADR-002 for Customer) all happen at this layer.

**Customer identity resolution:** The matching logic is fully specified in STM-001 section 4.3 – a four-tier hierarchy (ID number, deterministic → name + phone, high confidence → email + name, medium confidence → phone + name, low confidence, flagged for human validation). Silver implements this hierarchy as written. *Formal approval of ADR-002 remains pending Priya's POPIA legal review — this affects whether the approach is cleared for production use of real customer data, not whether the design itself is known. Design work proceeds on STM-001's specification; production go-live is gated on ADR-002 sign-off, per Step 13+ readiness criteria.*

**Data quality enforcement (FR-CCR-007):** Every constraint declared as informational-only throughout PDD-001 Section 5 is enforced here via dbt's built-in generic tests — no additional packages required for standard cases:

| PDD-001 Constraint | dbt Test | Notes |
|---|---|---|
| **PRIMARY KEY** | **not_null** + **unique** | dbt has no single "primary key" test; declare both on the surrogate key column |
| **UNIQUE** (single column) | **unique** | Direct match |
| **UNIQUE** (composite, e.g. **order_id + line_number**) | **dbt_utils.unique_combination_of_columns** | dbt-core's native **unique** test only checks one column; composite keys need the **dbt_utils** package |
| FOREIGN KEY | relationships | Confirms every child value exists in the parent table |
| Status/enum columns (**po_status, order_status**) | **accepted_values** | Confirms the column only ever holds the documented set of values |

Any test failure halts the pipeline before Gold is updated, per FR-CCR-007's acceptance criteria.

**Supplier Contract PDF structuring — distinct risk profile:** *This transformation differs structurally from the rest of Silver. Standard Silver transformations (cleansing, standardization, identity resolution) apply consistent, repeatable logic to structured Bronze data — dbt tests reliably catch their failure modes (NULLs, duplicates, out-of-range values). The pypdf+LLM extraction step applies an LLM to unstructured text on a one-off or on-request basis (new/amended contracts only, not a nightly recurring transform), and its dominant risk is different: a confidently wrong extraction (e.g. a misread contracted rate) that is structurally valid and will pass every standard dbt test, since nothing about it looks malformed. Standard data quality gates cannot catch this failure mode. Mitigating this requires a human spot-check step against source PDFs — sampling extracted values against the original document — rather than relying on automated testing alone. This is a build-phase (Step 14) design detail to resolve before this pipeline goes live, and it remains linked to ADR-003's own outstanding sign-off (Sipho Dlamini, per DM-001's outstanding items).*

# 5. Gold Layer Load Pattern

Gold layer loading logic differs by table type, following the classifications established in PDD-001 Section 5.2 and this section's dimension review.

**Fact table loading:**

- **Insert-only tables** (FactSales, FactFinance, FactInventorySnapshot) use a plain INSERT. Each row is created once and never revisited — there are no lifecycle or status columns for a later run to update. Protection against accidental duplicate loads (e.g. a pipeline rerun after a partial failure) is provided by the same dbt uniqueness tests declared in Section 4 (unique_combination_of_columns on each table's degenerate identifier or grain columns), not by the load mechanism itself. A duplicate load fails the pipeline via FR-CCR-007's test gate rather than being silently prevented at insert time.
- **Accumulating snapshot tables** (FactReplenishment, FactProcurement, FactContactHistory, FactSupplierContract) require a MERGE statement, matched on each table's declared unique constraint. When a match is found, the existing row's stage/status and duration columns are updated in place; when no match is found, a new row is inserted.

**Dimension loading – three distinct patterns:**

- **SCD Type 2** (DimProduct, DimCustomer, DimStore, DimEmployee, DimSupplier, DimAccount): when a tracked attribute changes, the existing row must **not** be overwritten — doing so would silently rewrite history for every fact row that referenced it. Instead, loading requires two steps: first, the existing current row is closed out (is_current = 0, effective_end_date set); second, a new row is inserted with the updated attribute values (is_current = 1, effective_start_date set, effective_end_date defaulted forward). A single MERGE cannot both update and insert against the same matched row, so this is implemented as two coordinated statements, not one. In practice, this pattern is commonly automated using dbt's built-in **snapshot** feature, which handles the close-out/insert-new logic natively rather than requiring hand-written MERGE logic — the exact implementation is a Step 14 build decision, but the two-step mechanic described here is what any implementation must achieve.
- **SCD Type 1** (DimCampaign): straightforward overwrite-in-place via MERGE, no history preserved. Appropriate here because a campaign's attributes are fixed once launched — an actual change in campaign dates or budget creates an entirely new campaign row, not a revision of the old one, so there is nothing SCD Type 2 would meaningfully protect.
- **Static / load-once** (DimDate, DimChannel): no ongoing extraction pipeline. DimDate is generated once as a full calendar range at platform setup; DimChannel (~7 rows) is manually maintained and updated only when a new channel is introduced. Neither requires a scheduled Gold load job.

# 6. Error Handling and Alerting

Satisfies FR-CCR-006: no silent failures — every pipeline error is detected, logged, and alerted with what failed, where, what data is affected, and recommended next steps, linked to a troubleshooting runbook.

**Routing:** All pipeline failures alert the data engineering team first — the team positioned to diagnose and resolve the technical fault. Domain owners (Sipho, Zanele, Rethabile, Priya, Johan) are not first-line recipients of routine pipeline alerts; they are escalated to when a failure meets defined escalation criteria — specifically, when a failure is not resolved within its layer's expected recovery window, or when a failure directly affects data, those owners are actively relying on for a time-sensitive decision. *Exact escalation thresholds (time-based, domain-based, or both) are a Step 14 build decision; to be confirmed once real failure patterns are observed during testing.*

**FR-FIN-005 exception — backdated posting alerts:** This requirement carries a stricter, named SLA distinct from FR-CCR-006's general pattern: backdated GL postings must alert **within 1 hour**, routed directly to Johan's team rather than following the standard engineer-first routing above, since this is a business-rule alert (a compliance/audit concern) rather than a technical pipeline failure. This exception must be demonstrated and explicitly approved by Johan during Step 14 testing before being considered satisfied — not just built and assumed correct.

**Runbook:** FR-CCR-006 requires every alert to link to a troubleshooting runbook. The runbook itself is not authored in this document — the specific failure modes it needs to cover are not yet known, and premature authoring risks documenting scenarios that don't match what testing actually surfaces. The runbook is a required Step 14 build artifact, developed iteratively as real failures are encountered and diagnosed during pipeline build and testing.

# 7. Reconciliation Framework

Satisfies FR-CCR-005: automated reconciliation checks at every layer boundary, daily variance reports, pipeline stops on failure.

Reconciliation runs at each layer boundary. **Bronze to Silver:** Bronze count = Silver count + logged rejected rows (rejected rows are written to a reject table with a reason, never dropped silently). **Silver to Gold:** the method varies by table type, not row counts alone:

- **Insert-only tables** (FactSales, FactInventorySnapshot): row-count reconciliation between Silver and Gold is a valid first check, since each Silver row loads to exactly one Gold row.
- **FactFinance and other monetary tables:** row counts are insufficient for audit purposes. Reconciliation compares **control totals** — e.g. sum of **debit_amount** in Silver versus sum of **debit_amount** in Gold for the same period — not individual row values. This satisfies the audit trail requirement (every reported number traceable to source, FR-CCR-001) without the performance cost of comparing every value cell-by-cell.
- **Accumulating snapshot tables** (FactReplenishment, FactProcurement, FactContactHistory, FactSupplierContract): row-count reconciliation is not meaningful here — a single business key (e.g. one purchase order) may generate multiple Bronze extracts across its lifecycle but only one current Gold row. Reconciliation instead confirms that Gold's current state matches the **most recent Silver state per business key**, validated via control totals (e.g. sum of **total_order_value** across all currently open orders) rather than per-key inspection.

**Failure isolation:** Each of the six business domains runs as an independently orchestrated ADF pipeline. A reconciliation failure halts only the affected domain's Gold load — it does not stop the platform's other five domains

from loading normally. This contains the blast radius of any single failure and allows unaffected domains' reporting to continue while the failing domain is investigated.

**Variance reporting:** Daily variance reports are generated regardless of pass/fail outcome, giving Johan's team (and other domain owners) visibility into reconciliation health over time, not just at the moment of failure. Sales-to-Finance reconciliation specifically depends on both domains' pipelines having completed for the day — SAP ECC's GL revenue entries arrive independently of RetailSphere's own Sales pipeline (BDR-001 confirms no source-system interface connects them), so the comparison itself, not either pipeline's load, is what requires both to finish first. The daily variance report for this specific comparison is timed accordingly, evaluated only after both domains report complete.

## 8. POPIA Opt-out Propagation

Satisfies FR-CUS-002 / FR-CCR-008: opt-out events propagated across all four marketing systems within four hours of the opt-out occurring — a non-negotiable POPIA legal obligation, distinct from Priya's Rule 4 same-day breach alert (Section 3), which addresses contact-cap breaches, not opt-out status.

Correction: Section 3's earlier reference to "FR-CUS-008" was an error — that requirement does not exist. The same-day breach alert reasoning in Section 3 correctly applies to Priya's Rule 4 (tied to FR-CUS-001's contact cap enforcement), not to opt-out propagation. This section addresses the separate FR-CUS-002/FR-CCR-008 requirement.

**Detection and propagation:** An opt-out can originate in any of the four systems — a Marketing Cloud unsubscribe link, a Salesforce CRM call center entry, a Meta ad preference change, or a GA4/website preference update. Rather than tracking origin and propagating only to the remaining three, the pipeline checks all four systems for new opt-out events and writes the opt-out status to all four — including the system where it originated — guaranteeing consistency without requiring origin-detection logic.

**Schedule:** This runs on its own dedicated hourly schedule, separate from the 4-hourly Meta/GA4 ingestion job established in Section 3. Hourly detection leaves sufficient time budget within the 4-hour maximum to confirm propagation across all four systems, satisfying FR-CUS-002's stated "hourly batch minimum" frequency requirement.

**Confirmation alerting:** Once propagation across all four systems is confirmed complete for a given opt-out event, a confirmation alert is sent to both Priya's team (as direct compliance evidence supporting her own POPIA audit trail) and the data engineering team (since an expected confirmation that fails to arrive is itself a signal of a pipeline issue, distinct from Section 6's failure-alert routing). This dual-routing differs deliberately from Section 6's standard pattern, where routine failures go to engineering first — here, the business owner is a first-class recipient because the alert itself is required regulatory evidence, not just operational status.

## 9. Monitoring

Monitoring leverages Azure Data Factory's native pipeline monitoring (portal-based run history, success/failure status, duration per run) as the primary mechanism — no custom monitoring platform is built for v1.0. Each of the six domain pipelines (per Section 7's isolation model) reports independently.

**Daily health check:** A single status metric — **6/6 domain pipelines completed** — serves as the first-glance signal of platform health. Anything less than 6/6 triggers investigation into which specific domain failed and why, using ADF's

native run history and the Section 6 alerting/runbook process to diagnose. This keeps the daily check simple and unambiguous: a partial success is still treated as requiring attention, not silently accepted as "mostly fine."

Note: this is a v1.0 baseline using native tooling only. As the platform matures past initial build (Steps 13+), a consolidated dashboard surfacing the 6/6 metric alongside reconciliation variance (Section 7) and alert volume (Section 6) may be worth building — flagged as a future enhancement, not a Step 10 design requirement.

## 10. Next Steps

The completion of DID-001 marks the end of Step 10 — Data Integration Design. The following items carry forward:

Immediate:

- Publish DID-001, update Notion step tracker and README Step 10 status.

**Step 11 — Metadata Strategy:** Define how lineage (FR-CCR-001), data dictionary, and column-level documentation are captured and maintained across Bronze, Silver, and Gold.

Outstanding items requiring resolution before build (Steps 13+)

- **ADR-002 formal sign-off** (Priya's POPIA legal review) — identity resolution *design* is fully specified (STM-001 section 4.3) and safe to build against, but production use of real customer data is gated on this sign-off completing first.
- **ADR-003 formal sign-off** (Sipho) — Supplier Contract PDF extraction approach, tied to the human spot-check quality process flagged in Section 4.
- **ADR-006 and ADR-007 sign-offs** — Marketing Attribution Logic and Contact Governance, both still IN PROGRESS per the ADR register; FR-MKT-002/003/004 and FR-CUS-001 depend on these before Marketing domain build.
- **FactContactHistory attempt-sequencing validation** (Section 5.2.5, PDD-001) — confirm against real Marketing Cloud send behaviour before the unique constraint is considered final.
- **Escalation criteria for pipeline failures** (Section 6) — exact thresholds (time-based, domain-based) for when a failure escalates beyond data engineering to domain owners; deferred to Step 14 once real failure patterns are observed.
- **Troubleshooting runbook** (Section 6) — authored iteratively during Step 14 build as real failure modes are encountered, not pre-written.
- **PDF extraction spot-check process** (Section 4) — human validation sampling approach for LLM-structured contract data needs to be designed before Step 14 build of that specific pipeline.
- **Monitoring dashboard** (Section 9) — native ADF monitoring is sufficient for v1.0; a consolidated dashboard is a future enhancement, not a current build requirement.

# 11. Document Change Log

| Version | Date | Author | Description |
|---|---|---|---|
| 1.0 - Draft | 4 September 2026 | Thozamile Mbalo | Initial DID-001 draft. Defines Bronze/Silver/Gold pipeline architecture pattern operationalizing ADR-005 (ADF orchestration). Covers ingestion patterns for 10 source systems including three non-standard cases (Meta/GA4 late-arriving data, Supplier Contract PDF extraction, Legacy SQL Server DW discovery-only); Silver-layer identity resolution referencing STM-001 section 4.3's already-specified matching hierarchy; Gold-layer load mechanics split by fact table type (insert-only vs. accumulating snapshot) and dimension SCD pattern (Type 1, Type 2, static); error handling and alerting (FR-CCR-006) with a named exception for FR-FIN-005's 1-hour SLA; reconciliation framework (FR-CCR-005) using control totals rather than row counts for monetary and accumulating snapshot tables; POPIA opt-out propagation (FR-CUS-002/FR-CCR-008) on a dedicated hourly schedule, distinct from the general Meta/GA4 4-hourly cadence; and native ADF monitoring with a single 6/6 domain health metric. Corrects an internal error: Section 3's original reference to a non-existent "FR-CUS-008" requirement, corrected to properly attribute the same-day breach alert to Priya's Rule 4 / FR-CUS-001. |
| 1.0 – Approved | 4 September 2026 | Reviewed by Nomvula Mahlangu, Chief Data Officer | Approved following review. Section 7 updated: clarified that Sales-to-Finance reconciliation timing (not pipeline execution) is what depends on both domains completing — confirmed via BDR-001's dependency map, which shows no source-system interface between POS/Shopify and SAP ECC's GL. Domain isolation model (Section 7) confirmed sound with no cross-domain execution dependency required. |
| 1.1 | 4 October 2026 | Thozamile Mbalo | Section 7 aligned with FR-CCR-005: reconciliation now explicitly covers Bronze to Silver as well as Silver to Gold, with legitimately rejected rows accounted for via a logged reject table. |

*— End of Document —*